CBSE BOARD EXAMINATION 2023

MATHEMATICS (STANDARD) / □□□□ (□□□□□)

Time Allowed: 3 Hours Maximum Marks: 80

General Instructions:

Section A consists of 8 questions of 1 mark each.
Section B consists of 4 questions of 2 marks each.
Section C consists of 4 questions of 3 marks each.
Section D consists of 4 questions of 5 marks each.

SECTION A (1 Mark each)

Q1. If two positive integers a and b are written as $a = x^3 y^2$ and $b = xy^3$, where x, y are prime numbers, ther
(□□□ □□ □□□□□□□□ □□□□□□□□□ $a = x^3 y^2$ □□ $b = xy^3$ □□□, □□ HCF(a, b) □□□□□□ □□□□□□□)

Q2. Find the zeroes of the quadratic polynomial $p(x) = x^2 - 3x - 4$. [1 mark]
(□□□□□□□□ □□□□□□ $p(x) = x^2 - 3x - 4$ □□ □□□□□□□ □□□□□□ □□□□□□□)

Q3. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$. [1 mark]
(□□□□□□□□ □□□□□□□ $2x^2 - 4x + 3 = 0$ □□ □□□□□□□□□□□ □□□□□□ □□□□□□□)

Q4. Which term of the AP: 2, 7, 12, ... is 47? [1 mark]
(□□□□□□□□ □□□□□□□ 2, 7, 12, ... □□ □□□□ □□ □□ 47 □□?)

Q5. Find the distance of point P(-6, 8) from the origin. [1 mark]
(□□□□□□ P(-6, 8) □□ □□□□ □□□□□□ □□ □□□□□ □□□□□□ □□□□□□□)

Q6. If sin theta + cos theta = sqrt(2) cos theta, then find the value of tan theta. [1 mark]
(□□□ sin theta + cos theta = sqrt(2) cos theta □□, □□ tan theta □□ □□□□ □□□□□□ □□□□□□□)

Q7. A card is drawn from a well-shuffled deck of 52 playing cards. Find the probability of getting a red king. [1 m
(52 □□□□ □□ □□□□□□□ □□□□ □□ □□ □□□□ □□□□□□□ □□□□□□ □□ □□□□□□□□□□□ □□□□□□ □□□□□□□)

Q8. If the mean of observations x, x+3, x+5, x+7, x+10 is 9, find the value of x. [1 mark]
(□□□ □□□□□□□□□□□ x, x+3, x+5, x+7, x+10 □□ □□□□□□ 9 □□, □□ x □□ □□□□ □□□□□□ □□□□□□□)

SECTION B (2 Marks each)

Q9. Prove that sqrt(5) is an irrational number. [2 marks]
(□□□□□ □□□□□ □□ sqrt(5) □□ □□□□□□□□ □□□□□□ □□□)

Q10. Solve the following pair of linear equations: 2x + 3y = 11 and 2x - 4y = -24. [2 marks]
(□□□□□ □□□□□□ □□□□□ 2x + 3y = 11 □□ 2x - 4y = -24 □□ □□ □□□□□□□)

Q11. Find the coordinates of the point which divides the line segment joining (-1, 7) and (4, -3) in ratio 2:3. [2 ma
(□□□□□□□ (-1, 7) □□ (4, -3) □□ 2:3 □□ □□□□□□ □□□ □□□□□□□ □□□□ □□□□ □□□□□ □□ □□□□

Q12. If tan(A + B) = sqrt(3) and tan(A - B) = 1/sqrt(3), find the values of A and B. [2 marks]
(□□□ tan(A + B) = sqrt(3) □□□ tan(A - B) = 1/sqrt(3) □□, □□ A □□ B □□□□□ □□□□□□)

SECTION C (3 Marks each)

Q13. Find the zeroes of polynomial 6x^2 - 3 - 7x and verify relationship with coefficients. [3 marks]
(□□□□□ 6x^2 - 3 - 7x □□ □□□□□□ □□□□□ □□□□□ □□□ □□□□□□□□ □□ □□□ □□□□□ □□□□□□□

Q14. If the sum of first 7 terms of an AP is 49 and 17 terms is 289, find sum of first n terms. [3 marks]
(□□□ AP □□ □□□□□ 7 □□□□ □□ □□□ 49 □□ 17 □□□□ □□ □□□ 289 □□, □□ □□□□□ n □□□□ □□ □

Q15. Prove that lengths of tangents drawn from an external point to a circle are equal. [3 marks]
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Q16. A solid cone of radius 1 cm stands on hemisphere of same radius. Height of cone is 1 cm. Find volume. [3
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SECTION D (5 Marks each)

Q17. A motor boat speed in still water is 18 km/h. Takes 1 hour more for 24 km upstream than downstream. Fi

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Q18. State and prove Basic Proportionality Theorem (Thales Theorem) for a triangle. [5 marks]

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Q19. From top of 7 m building, elevation of cable tower top is 60 deg, depression of foot is 45 deg. Find tower h

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Q20. Median of frequency distribution is 525. Find x and y if total frequency is 100. [5 marks]

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Q21. [BLURRED_FRAGMENT] What is the required condition for ??? [Needs Review]